Deep Learning - Lymphoma Data augmentation

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Abstract

The goal of this project is to evaluate the effects of data augmentation technique on the automated classification of lymphoma disease in the circumstance of limited training data. The proposed method tries to increase variety and quantity of training data by using standard augmentation technique like cropping patches and flipping to using affine and protective transformation in order to make the classifier model more robust and generalizable. Our experiments results shows that the augmented training data works quite well for increasing accuracy when dealing with medical image datasets having few samples to test on but many possible diagnoses alike themselves being hard or even impossible diagnose correctly only based on one instance alone.

1 Introduction

The diagnosis and planning of treatment are supported greatly by classification of medical images but usually the lacks of adequate labeled data lead to under performing machine learning models. To address this challenge, data augmentation has been introduced as an approach of artificially enlarging the dataset. This research presents a technique for producing fake pictures out from real pictures so as to add them to the training set. Our strategy is designed on deep learning architectures such as ResNet-50 coupled with different forms of data augmentation in order to enhance classification precision on medical images even when only few labelled instances are available.

2 Method Overview

Deep learning models have to be trained with data augmentation especially if the dataset is small or lacks variety. This particular implementation builds a strong data augmentation pipeline using *'transforms.Compose'* functionality in PyTorch.
The first step of the augmentation pipeline is converting input numpy arrays into PyTorch tensors through the custom *ArrayToTensor* transformation. Then, the following transformation are applied on order.

1. **Resized Crop**: crop a part from the original image and resize the cropped image to the specific resolution enhancing scale and translation invariance.

2. **Horizontal and Vertical Flip**: flip the image first horizontally then vertically, both are done with a set probability, making the model robust to orientation variation.

3. **Rotation**: randomly rotate the image by a set degree.

4. **Color Jittering**: brightness, contrast, saturation and hue of the image are randomly adjusted so that colour distribution diversity is increased.

5. **Grayscale Conversion**: the images is converted to grayscale at random with a set probability.

6. **Perspective and Affine Transformations**: applies, with set probability, prospective and affine transformation such as translation, scaling and sharing, thus applying spatial distortion.

7. **Erasing**: erase at random a rectangular region of the image, making the model robust to partial information.

8. **Normalizing**: normalizes the image to the ImageNet mean and standard deviation

3 Training

The augmented dataset is then used to train a ResNet-50 model which is a CNN architecture designated specifically for image classification. It consist of 50 layers, including convolutional, pooling, fully connected and shortcut connections known as residual connections. This block uses a 1×1 convolutions, known as a “bottleneck”, which reduces the number of parameters and matrix multiplications. This enables much faster training of each layer and mitigates the vanishing gradient problem.

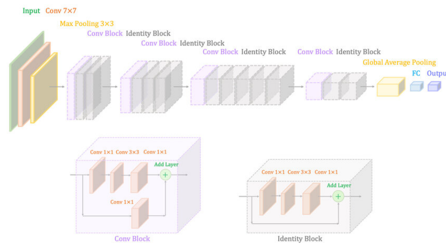

Figure 1: ResNet-50 architecture

4 Results

The presented model has obtained and accuracy of 85.56% on the 374 test images. The following graph shows the loss on the test.

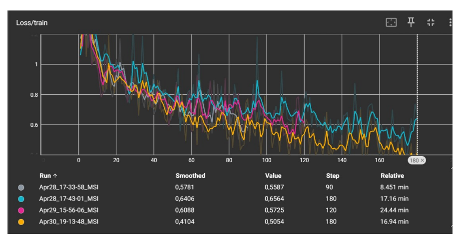

Figure 2: Test loss

5 Conclusion

We observed good results on the augmented dataset, achieving an accuracy of 85.56% on the test set; however the model trained on the non-augmented dataset yielded a better accuracy of 85.83%. Although the gap in accuracy of the two trained models is small, augmenting the dataset potentially enhance the ability of the model to generalize and withstand some different variations.

In summary, our results suggest that data augmentation can enrich the diversity of the training data and allow the model to have capability of grasping resilient features. Feature works could expand some other augmentation approaches and evaluate their impact on model performance across different medical dataset.